Cloud-Based Exam Question Bank with AI-Assisted Exam Generation

Technical Progress Report | Step 1: Foundation (sub-steps 1 to 4)

3	13	8	55/55
migrations 5 CHECK constraints	API operations in 9 route files	secondary indexes 6 foreign keys	end-to-end API checks passing

Item	Detail
Report date	19 October 2026
Repository	`github.com/sannnntf123-pixel/mycapstone`
Stack	Next.js 15.5 (App Router, TypeScript) | PostgreSQL 16 | Prisma 6.19 | JWT (jose) | Zod
Users	Instructors only. Instructors author and approve questions and (in a later step) generate exams.
Status	Project setup, schema, REST API, security hardening and demo seed data are complete and verified. Automated test suite, app Dockerfile and README are next.

1. Executive summary

This report documents the first delivered increment of the capstone: the back-end foundation for a question bank used by **instructors**. Instructors author exam questions; every question passes an explicit instructor approval step before it enters the bank; and each question can be kept private or shared with other instructors. Later increments will add AI question generation and AI exam assembly, both for instructors. The schema already carries the fields those features need, so they can be added without restructuring the data model.

Scope change (19 Oct 2026): the system no longer has a Student role. Student exam-taking was removed from the plan and the role was dropped from the schema with a dedicated migration (Section 4.6). The system is instructor-only, and every endpoint except registration and login requires an authenticated instructor.

The implementation is a Next.js 15 application exposing a JSON REST API through App Router route handlers, backed by PostgreSQL through the Prisma ORM. Authentication is stateless (signed JWT bearer tokens) and gated by an invite code; authorization for questions is row-level: the visibility rule is defined in exactly one function and every question query is composed with it. All 55 automated end-to-end checks pass against the running server and a real database.

Hardening (sub-step 3): the database now enforces the business rules itself (CHECK constraints and a composite foreign key), failed logins and invite-code guesses are rate-limited, login timing no longer reveals which emails exist, configuration is validated at startup, and malformed requests return 400 instead of 500. The test suite grew to 55 checks and caught a real IP-spoofing weakness during development (Section 12.1).

1.1 Scope and status

#	Sub-step (Step 1)	Status	Evidence
1	Project setup: Next.js, Prisma, PostgreSQL in Docker, schema and initial migration	**Done**	Migration 20261016091130_init
2	REST API: auth, curricula/chapters, questions (create, list/filter, get, approve, reject, bank)	**Done**	Live checks; tsc + eslint clean
2b	Scope change: remove Student role, instructor-only system	**Done**	Migration 20261019100000_remove_student_role
3	Guard review and hardening: DB constraints, rate limiting, timing-safe login, startup config validation	**Done**	Migration 20261019120000_db_integrity_checks; 55/55 checks
4	Demo seed data: 3 instructors, 2 curricula, 7 chapters, 25 questions in every state	**Done**	npm run db:seed; check:seed 17/17
5	Automated test suite (approval + privacy) on an isolated test DB	**Next**	scripts/smoke-test.mjs to be ported
6	App Dockerfile, full docker-compose, README	To do	DB compose file exists

Out of scope for this step: AI question generation, exam generation and any user interface. Student exam-taking has been removed from the project.

2. Technology stack

Layer	Choice (version)	Rationale
Runtime / framework	Next.js 15.5.28, React 19.1, TypeScript 5, Node 23 (dev)	One codebase serves the API now and the UI later. Route handlers are standard Web Request/Response functions, deployable to Vercel (serverless) or any container host.
Database	PostgreSQL 16 (postgres:16-alpine)	Data is strongly relational (curriculum > chapter > question > user). Native enum types, JSON columns, mature managed offerings on every cloud (RDS, Cloud SQL, Azure, Neon, Supabase).
ORM / migrations	Prisma 6.19.3	Declarative schema is both documentation and source of truth; versioned SQL migrations and a fully typed client, so a typo in a column name is a compile error.
Authentication	jose 6.2 (JWT HS256), bcryptjs 3.0 (cost 10)	Stateless tokens scale horizontally with no session store. jose runs on both the Node and Edge runtimes. Chosen over NextAuth because its Credentials provider makes server-side RBAC and API testing awkward.
Validation	Zod 3.25	Every body and query string is parsed at the boundary; types are inferred from schemas, so validation and TypeScript types cannot drift apart.
Containers	Docker via Colima, docker-compose	Reproducible database locally; same image family used in the cloud.

3. Architecture

The code is organised in strict layers. Each layer only calls the one beneath it, which keeps route handlers trivial and concentrates every business rule in a domain service that can be unit-tested without HTTP.

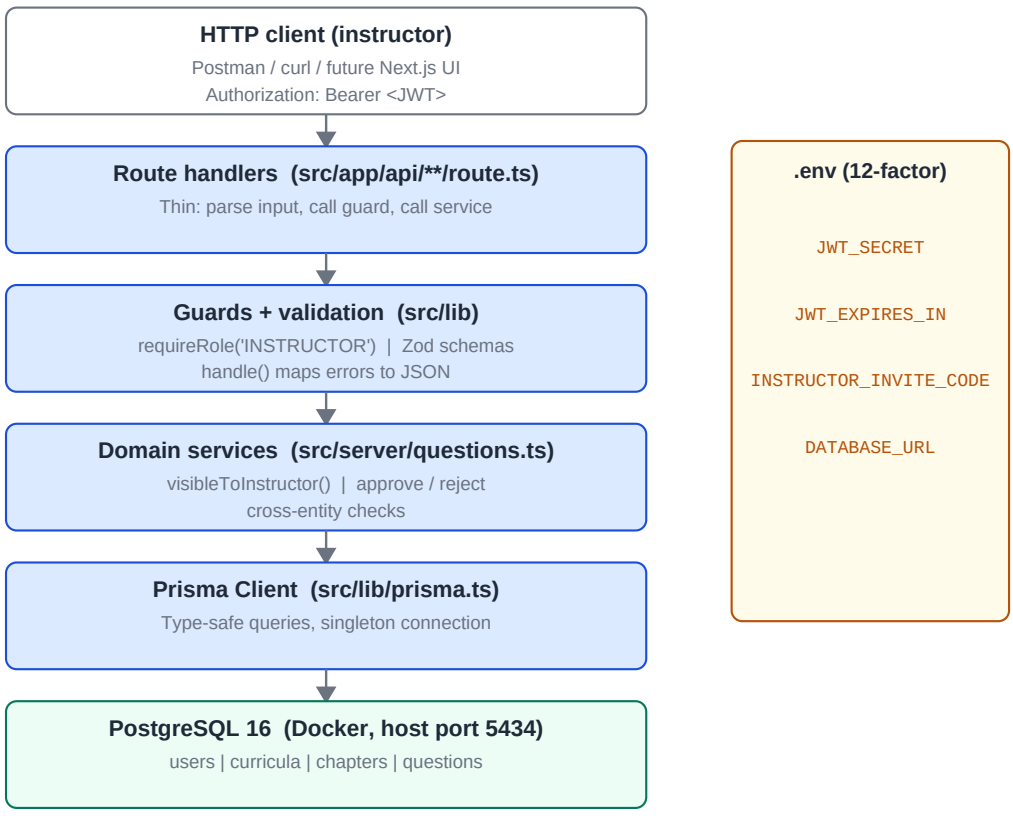

Figure 1 - Layered architecture and request path

3.1 Folder structure

```
my-capstone/
|-- prisma/
|   |-- schema.prisma                data model (source of truth)
|   |-- migrations/                  init, remove_student_role, db_integrity_checks
|   `-- seed.ts                      idempotent demo data (npm run db:seed)
|-- src/
|   |-- app/api/                     HTTP layer - thin route handlers
|   |   |-- auth/register|login|me/route.ts
|   |   |-- curricula/route.ts, curricula/[id]/chapters/route.ts
|   |   `-- questions/route.ts, bank/route.ts,
|   |       [id]/route.ts, [id]/approve/route.ts, [id]/reject/route.ts
|   |-- instrumentation.ts           validates config at server startup
|   |-- lib/                         cross-cutting infrastructure
|   |   |-- env.ts                   Zod-validated environment
|   |   |-- prisma.ts                singleton PrismaClient
|   |   |-- auth.ts                  hashing, JWT, requireUser / requireRole
|   |   |-- rate-limit.ts            failed-attempt counters (429)
|   |   |-- errors.ts                HttpError, handle(), readJson()
|   |   `-- validation.ts            Zod schemas for every input
|   `-- server/questions.ts          privacy filter + approval workflow
|-- scripts/smoke-test.mjs           end-to-end API checks (npm run smoke)
|-- scripts/check-seed.mjs           verifies seeded data per instructor (npm run check:seed)
|-- docs/                            this report + its generator
|-- docker-compose.yml   .env.example   PROCESS.md
```

3.2 Request lifecycle

- The route handler is wrapped in `handle()`, so any thrown error becomes a JSON response with the right status code.
- `requireRole(req, 'INSTRUCTOR')` verifies the bearer JWT signature and expiry, then **reloads the user from the database** and checks the role.
- The request body or query string is parsed by a Zod schema; failure produces HTTP 400 with per-field messages.
- The domain service applies business rules (visibility, chapter-belongs-to-curriculum, status transitions) and queries via Prisma.
- The handler returns `NextResponse.json(...)` with 200/201.

Extension points for later steps: `src/server/ai.ts`, `src/server/exams.ts` and `src/app/api/exams/**` plug in at the same layers; UI pages go under `src/app/`.

4. Database design

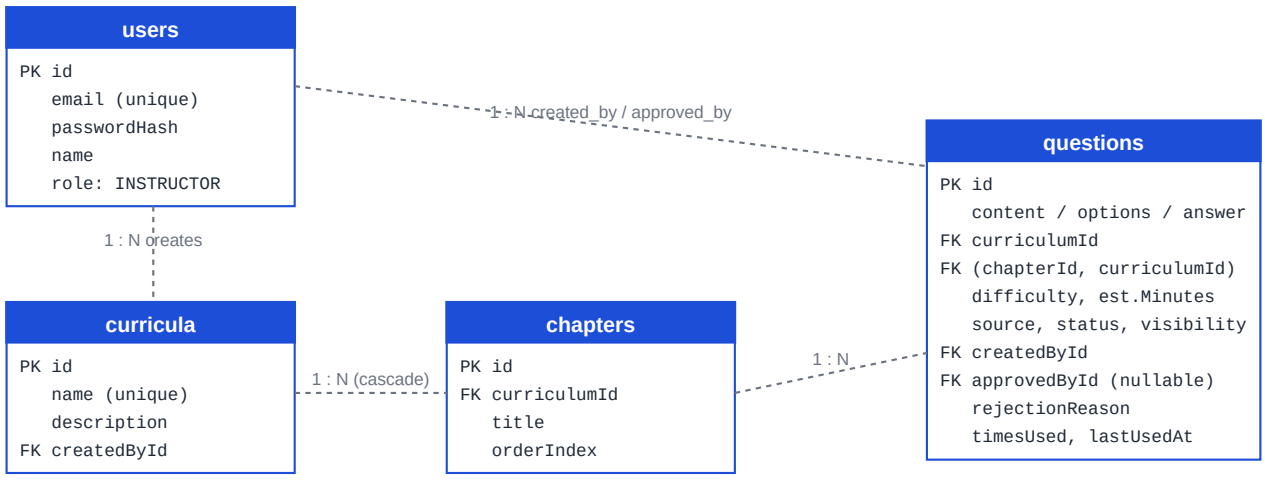

Figure 2 - Entity-relationship overview

Primary keys are `cuid` strings rather than auto-increment integers: they are non-guessable (no enumeration of other instructors' questions by incrementing an id), can be generated without a database round-trip, and are safe to merge across environments. Table names are mapped to snake-case plurals via `@@map`. All tables carry `createdAt` and an automatically maintained `updatedAt`.

4.1 Enumerated types (native PostgreSQL ENUMs)

Type	Values	Used by
Role	INSTRUCTOR	users.role (default INSTRUCTOR). Kept as an enum so further roles (e.g. Admin) can be added by migration without touching the auth layer.
QuestionStatus	PENDING, APPROVED, REJECTED	questions.status (default PENDING)
Visibility	PRIVATE, SHARED	questions.visibility (default PRIVATE)
QuestionSource	MANUAL, AI	questions.source (default MANUAL; AI reserved for a later step)

4.2 Table: questions

Column	Type	Null	Notes
id	text (cuid)	no	Primary key
content	text	no	Question stem
options	jsonb	yes	Array of choices for MCQ; null for open questions. API enforces answer is one of them.
answer	text	no	Correct answer / model solution
curriculumId	text FK	no	-> curricula.id, ON DELETE RESTRICT
chapterId	text FK	no	Composite FK (chapterId, curriculumId) -> chapters(id, curriculumId): the DB guarantees the chapter belongs to the curriculum
difficulty	integer	no	1-5 (Zod + DB CHECK)
estimatedTimeMinutes	integer	no	1-600 (Zod + DB CHECK); input for exam-duration planning
source	QuestionSource	no	Forced to MANUAL by the create endpoint
createdById	text FK	no	-> users.id, ON DELETE RESTRICT; the owner
status	QuestionStatus	no	Forced to PENDING on create
approvedById	text FK	yes	-> users.id, ON DELETE SET NULL; set only on approval
approvedAt	timestamp(3)	yes	Set exactly when status = APPROVED (DB CHECK)
rejectionReason	text	yes	Optional reviewer comment on rejection
visibility	Visibility	no	PRIVATE by default (privacy-safe default)
timesUsed	integer	no	Default 0, never negative (DB CHECK); incremented by exam generation later
lastUsedAt	timestamp(3)	yes	Lets exam generation prefer less recently used questions
createdAt / updatedAt	timestamp(3)	no	Audit timestamps

4.3 Tables: users, curricula, chapters

Table	Columns	Constraints
users	id, email, passwordHash, name, role, createdAt, updatedAt	email UNIQUE. Only a bcrypt hash is stored, never the password.
curricula	id, name, description?, createdById, createdAt, updatedAt	name UNIQUE; createdById -> users (RESTRICT)
chapters	id, curriculumId, title, orderIndex, createdAt, updatedAt	UNIQUE(curriculumId, orderIndex), UNIQUE(curriculumId, title), UNIQUE(id, curriculumId); orderIndex > 0 (CHECK); curriculumId -> curricula ON DELETE CASCADE

4.4 Indexes and why each exists

Index	Columns	Serves
questions_curriculumId_chapterId_idx	(curriculumId, chapterId)	Filtering by curriculum and chapter; the primary access path for exam assembly
questions_status_visibility_idx	(status, visibility)	Bank queries (status = APPROVED) combined with the shared-visibility predicate
questions_createdById_idx	(createdById)	The 'own questions' branch of the privacy filter and ?mine=true
questions_difficulty_idx	(difficulty)	Difficulty filtering / difficulty-balanced exam generation
users_email_key	UNIQUE(email)	Login lookup and duplicate-registration prevention
curricula_name_key	UNIQUE(name)	Prevents duplicate curricula
chapters_..._orderIndex_key / _title_key	UNIQUE per curriculum	Stable chapter ordering; no duplicate chapter titles
chapters_id_curriculumId_key	UNIQUE(id, curriculumId)	Target of the composite foreign key from questions

4.5 Referential-integrity decisions

- **Deleting a curriculum cascades to its chapters**, but is **blocked** (RESTRICT) while any question references it, so approved bank content can never disappear silently.
- **Deleting a user is blocked** while they own questions or curricula. If an approver is removed, `approvedById` becomes NULL (SET NULL) while `approvedAt` keeps the historical fact.
- Denormalising `curriculumId` onto questions (it is derivable through the chapter) is deliberate: it makes the most common filter a single-column index lookup with no join. The risk of denormalisation (the two copies disagreeing) is removed by the composite foreign key (Section 4.7).

4.6 Migration history

Migration	Change
20261016091130_init	Creates the four tables, four enum types, eight secondary indexes and six foreign keys.
20261019100000_remove_student_role	Deletes accounts with role STUDENT, then recreates the Role enum with INSTRUCTOR only (PostgreSQL cannot drop an enum value in place) and sets the default to INSTRUCTOR. Safe: students could never own curricula or questions, so no rows are orphaned.

```
DELETE FROM "users" WHERE "role" = 'STUDENT';
CREATE TYPE "Role_new" AS ENUM ('INSTRUCTOR');
ALTER TABLE "users" ALTER COLUMN "role" DROP DEFAULT;
ALTER TABLE "users" ALTER COLUMN "role" TYPE "Role_new" USING ("role"::text::"Role_new");
ALTER TYPE "Role" RENAME TO "Role_old";
ALTER TYPE "Role_new" RENAME TO "Role";
DROP TYPE "Role_old";
ALTER TABLE "users" ALTER COLUMN "role" SET DEFAULT 'INSTRUCTOR';
```

After applying it, `prisma migrate diff` between the live database and `schema.prisma` produced an empty migration, confirming the database and schema are in sync.

Migration	Change
20261019120000_db_integrity_checks	Adds the composite foreign key and five CHECK constraints described in Section 4.7. Before applying it, the existing rows were queried and none violated the new rules.

4.7 Rules enforced by the database itself (defence in depth)

Every rule below was already validated by the API. Enforcing it a second time in PostgreSQL means that a future bug, a new code path (for example the AI generator writing questions), or a manual SQL edit still cannot store invalid data. Prisma's schema language cannot express CHECK constraints, so they live in a hand-written migration.

Constraint	Rule	Why
questions_chapterId_curriculumId_fkey	(chapterId, curriculumId) -> chapters(id, curriculumId)	A question's chapter must belong to the question's curriculum; makes the denormalised curriculumId safe
questions_difficulty_range	difficulty BETWEEN 1 AND 5	Exam generation relies on the 1-5 scale
questions_estimated_time_range	estimatedTimeMinutes BETWEEN 1 AND 600	Exam duration sums must be meaningful
questions_times_used_non_negative	timesUsed >= 0	Usage statistics can never go negative
questions_approval_consistency	(status = 'APPROVED') = (approvedAt IS NOT NULL)	Approval date exists exactly for approved questions. approvedById is excluded because it becomes NULL if the approver's account is deleted
chapters_order_index_positive	orderIndex > 0	Chapter numbering starts at 1

```
ALTER TABLE "questions" ADD CONSTRAINT "questions_chapterId_curriculumId_fkey"
  FOREIGN KEY ("chapterId", "curriculumId") REFERENCES "chapters"("id", "curriculumId");
ALTER TABLE "questions" ADD CONSTRAINT "questions_approval_consistency"
  CHECK (("status" = 'APPROVED') = ("approvedAt" IS NOT NULL));
```

The test script verifies each rule by running SQL directly inside the database container, bypassing the API, and checking that PostgreSQL rejects the write with the expected constraint name (Section 11).

5. Authentication and access control

5.1 Registration and login

- Passwords are hashed with **bcrypt, cost factor 10** (salted, deliberately slow); length 8 characters to **72 bytes**. bcrypt silently ignores everything after 72 bytes, so longer passwords are rejected rather than giving a false sense of security.
- **Every registration requires an invite code** equal to the `INSTRUCTOR_INVITE_CODE` environment variable (missing code: 400, wrong code: 403). Without this, anyone on the internet could create an account and read the shared question bank. If the variable is not set, registration fails closed.
- The role is **always set server-side** to INSTRUCTOR; a `role` field sent by the client is ignored.
- Login returns the same *'Invalid email or password'* (401) for an unknown email and a wrong password, and **takes the same time**: for an unknown email the password is still compared against a fixed dummy bcrypt hash. Measured: 56 ms vs 55 ms average. Without this, the faster answer for unknown emails would reveal which accounts exist.
- Emails are normalised to lower case before storage and lookup.

5.2 Token design

Property	Value
Format / algorithm	JWT, HS256 (HMAC-SHA256) signed with JWT_SECRET (random 256-bit, generated with openssl)
Claims	`sub` = user id, `role`, `iat`, `exp`
Lifetime	JWT_EXPIRES_IN, default 8h (a teaching day)
Transport	`Authorization: Bearer <token>` header (no cookies, therefore no CSRF surface for the API)
Verification	jose.jwtVerify with **algorithms pinned to HS256** (a token claiming another algorithm, e.g. 'none', is rejected); then the user row is reloaded
Secret strength	JWT_SECRET must be at least 32 characters, checked at server startup (Section 5.5)

Why reload the user on every request? The token's `role` claim is informational only; authorization uses the role read from the database. A deleted account or a changed role therefore takes effect on the next request instead of when the token expires. The cost is one primary-key lookup per request.

```
export async function requireRole(req: Request, ...roles: Role[]) {
  const user = await requireUser(req);                // 401 if missing/invalid/expired
  if (!roles.includes(user.role)) throw forbidden();   // 403 if wrong role
  return user;
}
```

Every protected route calls `requireRole(req, 'INSTRUCTOR')` explicitly, even though INSTRUCTOR is currently the only role. This keeps each endpoint's permission visible in its own code and means that adding a new role later cannot silently grant it access to existing endpoints.

5.3 Permission matrix

Operation	Anonymous	Instructor
Register (with invite code) / login	yes	-
GET /api/auth/me	no (401)	**yes**
List curricula and chapters	no (401)	**yes**
Create curriculum / chapter	no (401)	**yes**
Create question	no (401)	**yes**
List / filter / get questions	no (401)	own + shared only
View question bank	no (401)	approved, own + shared
Approve / reject	no (401)	pending questions visible to them
Generate exams (later step)	no (401)	planned

Audit (sub-step 3): all 13 operations were checked against this matrix. Every handler is wrapped in `handle()`, every protected handler calls `requireRole(req, 'INSTRUCTOR')` (`/auth/me` was changed from a plain login check to the explicit role check), and every request body goes through `readJson()` and a Zod schema.

5.4 Rate limiting (brute-force protection)

Target	Counter key	Rule
Login	`login:<email>`	After 5 failed logins for an account within 15 minutes, every login attempt for it returns 429 with a Retry-After header, even with the correct password. A successful login resets the counter.
Registration	`register:<client IP>`	After 5 wrong invite codes within 15 minutes, registration returns 429, even with the correct code (otherwise a correct guess would still be confirmed).

- Limits are configurable with `RATE_LIMIT_MAX_ATTEMPTS` and `RATE_LIMIT_WINDOW_MINUTES`.

- **Login is limited per account, not per IP.** An attacker who rotates IP addresses still gets only 5 guesses per account per window. Trade-off: an attacker can deliberately lock a known account for 15 minutes (a nuisance, not a breach).
- **Client IPs are only trusted behind a proxy.** `X-Forwarded-For` is set by the client unless a proxy (Vercel, nginx) overwrites it. With `TRUST_PROXY=false` (default) all invite-code attempts share one counter; with `TRUST_PROXY=true` the proxy-appended last hop is used.
- The counters live in server memory behind a small `AttemptStore` interface: correct for one instance; a multi-instance deployment would plug in a shared store such as Redis without changing the routes.

5.5 Fail-fast configuration

All environment variables are validated with Zod in `src/lib/env.ts`, called from `src/instrumentation.ts`, which Next.js runs once at server startup. A missing or weak value stops the server immediately with a clear message, instead of failing on the first request in production. Verified by starting the server with an 8-character secret:

```
Error: An error occurred while loading instrumentation hook: Invalid environment configuration:
  - JWT_SECRET: JWT_SECRET must be at least 32 characters (use: openssl rand -base64 32)
```

6. Privacy model (row-level authorization)

Rule: an instructor may see a question if and only if they created it, or its visibility is SHARED. The rule lives in a single function:

```
export function visibleToInstructor(userId: string): Prisma.QuestionWhereInput {
  return { OR: [{ createdById: userId }, { visibility: "SHARED" }] };
}

// every query composes the rule with the caller's filters using AND:
const where = { AND: [ visibleToInstructor(user.id),
                       { curriculumId, chapterId, difficulty, status, visibility, ... } ] };
```

Because the rule is combined with `AND`, no client-supplied filter can widen visibility. For example, `?visibility=PRIVATE` returns only the caller's own private questions, never another instructor's (verified in Section 11). The rule is applied inside the database query rather than by filtering results in memory, so hidden rows never leave the database and pagination counts stay correct.

Enforcement point	Mechanism
GET /api/questions, /api/questions/bank	listQuestions() composes visibleToInstructor() into findMany and count
GET /api/questions/:id	getVisibleQuestion(): findFirst with id AND visibility rule; else 404
approve / reject	Both call getVisibleQuestion() first, so you cannot review what you cannot see
Anonymous callers	requireRole(req, 'INSTRUCTOR') on every route; 401 before any query runs

404 rather than 403 is returned for another instructor's private question, so the API does not even confirm that the question exists.

7. Approval workflow

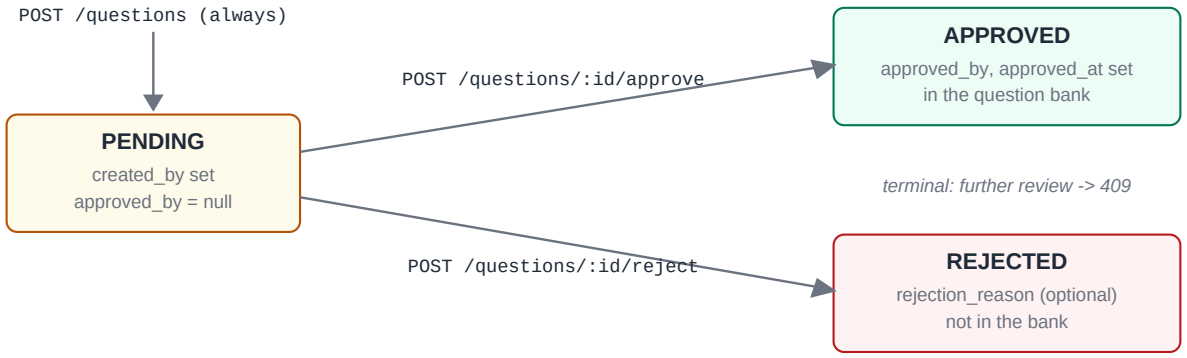

Figure 3 - Question status state machine

Transition	Preconditions	Fields written
create -> PENDING	Instructor; chapter belongs to curriculum; MCQ answer within options	status=PENDING, source=MANUAL, createdById=caller (all server-controlled, client cannot override)
PENDING -> APPROVED	Instructor who can see the question; status still PENDING	approvedById=caller, approvedAt=now(), rejectionReason=null
PENDING -> REJECTED	Same as approve; optional reason	rejectionReason=reason
Reviewed -> any	Not allowed (APPROVED and REJECTED are final)	HTTP 409 Conflict

7.1 Who may approve

Any instructor who can see the question. Shared questions therefore get peer review from any colleague; a private question can only be approved by its owner (nobody else can see it). Whether an owner should be allowed to approve their *own shared* question is an open design question (Section 12).

7.2 Concurrency safety

Two instructors could press 'approve' at the same moment. A naive read-check-write would let both succeed and the last write would overwrite `approvedById`. The update is instead made **conditional on the current status**, which PostgreSQL evaluates atomically:

```
const result = await prisma.question.updateMany({
  where: { id, status: "PENDING" },     // compare-and-set
  data,
});
if (result.count === 0) throw conflict("Question was reviewed by someone else");   // 409
```

7.3 The question bank

The bank is defined as **status = APPROVED**, intersected with the visibility rule. `GET /api/questions/bank` forces `status: 'APPROVED'` server-side, overriding any status in the query string; this is the endpoint the exam generator will draw from in a later step.

8. Validation and error handling

Schema	Rules
registerSchema	email valid + lower-cased; password >= 8; name non-empty; inviteCode required (non-empty)
createQuestionSchema	content, answer non-empty; options: array of >= 2 non-empty strings (optional); difficulty int 1-5; estimatedTimeMinutes int 1-600; visibility PRIVATE|SHARED (default PRIVATE)
listQuestionsQuery	query strings coerced to numbers; status/visibility enums; page >= 1; pageSize 1-100 (default 20)
createChapterSchema	title non-empty; orderIndex positive int
rejectQuestionSchema	reason optional; body itself optional

Unknown fields such as `role`, `status` or `createdById` are stripped by Zod, so clients cannot mass-assign protected fields (verified in Section 11).

Request bodies are read with `readJson()`: malformed JSON or a missing body now returns **400** instead of crashing with a 500. Before sub-step 3, the reject endpoint silently treated malformed JSON as 'no reason given'; it now rejects it too, while a genuinely empty body is still allowed there because the reason is optional.

Status	Meaning in this API
200 / 201	Success / resource created
400	Validation failed (with per-field details), malformed JSON, or inconsistent input (e.g. chapter not in curriculum)
401	Missing, invalid or expired token; bad login
403	Wrong role; invalid invite code
404	Not found, or not visible to the caller
409	Duplicate (email, curriculum, chapter) or invalid status transition / concurrent review
429	Too many failed logins or invite-code attempts; Retry-After header gives the seconds to wait
500	Unexpected error; logged server-side, generic message to client (no stack traces leak)

```
HTTP/1.1 400 Bad Request
{ "error": "Validation failed",
  "details": { "difficulty": ["Number must be less than or equal to 5"] } }
```

9. API reference

Method	Path	Access	Description
POST	/api/auth/register	public + invite	Create instructor account; returns { user, token } (201)
POST	/api/auth/login	public	Returns { user, token }
GET	/api/auth/me	instructor	Current user
GET	/api/curricula	instructor	All curricula with ordered chapters
POST	/api/curricula	instructor	Create curriculum { name, description? }
GET	/api/curricula/:id/chapters	instructor	Chapters of a curriculum, ordered
POST	/api/curricula/:id/chapters	instructor	Create chapter { title, orderIndex }
GET	/api/questions	instructor	List visible questions; filters + pagination
POST	/api/questions	instructor	Create question (PENDING)
GET	/api/questions/bank	instructor	Approved visible questions only
GET	/api/questions/:id	instructor	Single visible question
POST	/api/questions/:id/approve	instructor	Approve a pending question
POST	/api/questions/:id/reject	instructor	Reject; optional { reason }

List filters: `curriculumId, chapterId, difficulty, status, visibility, mine=true, page, pageSize`. Results are ordered newest first and returned as `{ items, total, page, pageSize }`.

9.1 Example: register and create a question

```
POST /api/auth/register
{ "email": "alice@uni.edu", "password": "********", "name": "Alice", "inviteCode": "<code>" }
201 Created  { "user": { "id": "cmg...", "role": "INSTRUCTOR", ... },
               "token": "eyJhbGciOiJIUzI1NiJ9..." }

POST /api/questions
Authorization: Bearer eyJhbGciOiJIUzI1NiJ9...
{ "content": "What is 1 + 1?", "options": ["1","2","3"], "answer": "2",
  "curriculumId": "cmg...", "chapterId": "cmg...", "difficulty": 1,
  "estimatedTimeMinutes": 2, "visibility": "SHARED" }

201 Created   (abridged)
{ "id": "cmg...", "status": "PENDING", "source": "MANUAL", "visibility": "SHARED",
  "approvedBy": null, "approvedAt": null, "timesUsed": 0,
  "chapter": { "title": "Algebra", "orderIndex": 1 }, "createdBy": { "name": "Alice" } }
```

9.2 Example: approve

```
POST /api/questions/cmg.../approve          -> 200 { ..., "status": "APPROVED",
                                                     "approvedBy": { "name": "Bob" },
                                                     "approvedAt": "<ISO-8601 timestamp>" }
POST /api/questions/cmg.../approve (again)  -> 409 { "error": "Question is already approved" }
```

10. Configuration and environment

Variable	Purpose	Example / default
DATABASE_URL	PostgreSQL connection string	`postgresql://capstone:***@localhost:5434/question_bank`
JWT_SECRET	HMAC key for signing tokens (min 32 chars)	random, from `openssl rand -base64 32`
JWT_EXPIRES_IN	Token lifetime	8h
INSTRUCTOR_INVITE_CODE	Required to register (min 8 chars)	set per deployment
RATE_LIMIT_MAX_ATTEMPTS / _WINDOW_MINUTES	Brute-force limits	5 / 15
TRUST_PROXY	Trust X-Forwarded-For from a fronting proxy	false (true on Vercel / behind nginx)
SEED_PASSWORD	Password of the demo instructors (dev only)	set locally; min 8 chars
DB_PORT, POSTGRES_*	docker-compose overrides	5434, capstone, question_bank

All configuration is read from the environment (12-factor), so the same build runs locally and in the cloud with different values. `.env` is git-ignored; `.env.example` is committed as the template.

npm script	Command
dev	next dev --turbopack -p 3100
db:up / db:down	docker-compose up -d / docker-compose down
db:migrate	prisma migrate dev
db:studio	prisma studio (visual DB browser)
db:seed	prisma db seed (runs prisma/seed.ts with tsx)
smoke	node scripts/smoke-test.mjs (end-to-end checks against a running server)
check:seed	node --env-file=.env scripts/check-seed.mjs (per-instructor visibility of the seed data)

npm script	Command
typecheck / lint	tsc --noEmit / eslint

```
npm install
cp .env.example .env          # set JWT_SECRET and INSTRUCTOR_INVITE_CODE
npm run db:up                 # Postgres on localhost:5434
npm run db:migrate            # apply migrations, generate client
npm run db:seed               # demo instructors, curricula and questions
npm run dev                   # API on http://localhost:3100
npm run smoke                 # in a second terminal
```

10.1 Demo data (seed)

`prisma/seed.ts` fills a development database with realistic data covering every state the system supports. It is **idempotent** (users, curricula and chapters are upserted by their unique keys; a question is only created if its owner does not already have one with the same text), never deletes rows, and **refuses to run when NODE_ENV=production**, so demo accounts cannot reach a live deployment. Running it twice was verified to create 25 questions and then 0.

Item	Seeded content
Instructors	Dr. Lina Nasser (lina.nasser@example.com), Dr. Omar Khalil (omar.khalil@example.com), Dr. Sara Haddad (sara.haddad@example.com); password from SEED_PASSWORD. Fictional people on the reserved example.com domain.
Mathematics 101	Algebra, Functions, Limits, Derivatives (owner: Lina)
Introduction to Programming	Variables and Types, Control Flow, Functions (owner: Omar)
Questions (25)	15 approved, 7 pending, 3 rejected (each with a reason); 19 shared, 6 private; difficulty 1-5; 1-20 minutes; 8 multiple-choice; 8 approved questions with usage history (timesUsed, lastUsedAt)
AI-generated	2 questions with source = AI, both still PENDING: AI output enters the same approval workflow as manual questions
Peer review	Shared questions are approved by a different instructor; private ones by their owner (the only person who can see them)

`scripts/check-seed.mjs` logs in as each demo instructor and compares what the API returns with what the privacy rule says they should see. **17 of 17 checks passed**:

Instructor	Questions visible (own + shared)	Question bank (approved)
Dr. Lina Nasser	21	14
Dr. Omar Khalil	22	14
Dr. Sara Haddad	20	13

Each instructor sees a different number because each sees their own private questions but not anyone else's. The script also confirms that no instructor ever receives another instructor's private question, that both AI questions are still pending, that every approved question records its approver and date, and that every rejected question has a reason.

11. Verification

Static checks: `tsc --noEmit` and `eslint` report zero errors; `prisma migrate diff` shows no drift between database and schema. **Dynamic checks:** `scripts/smoke-test.mjs` registers fresh instructors and exercises every endpoint over HTTP against the running server and the real PostgreSQL database; the database-rule checks run SQL directly inside the container. Result after sub-step 3: **55 of 55 checks passed**.

Area	Check	Result
Auth	Register without invite code -> 400; wrong invite code -> 403	**PASS**

Area	Check	Result
Auth	Client-supplied role is ignored; account is INSTRUCTOR	**PASS**
Auth	Register two instructors; duplicate email -> 409	**PASS**
Auth	Login returns token; wrong password -> 401	**PASS**
Auth	/auth/me returns current user; garbage token -> 401	**PASS**
Access	No token on /auth/me, /curricula, /questions, /questions/bank -> 401 (4 checks)	**PASS**
Curricula	Create curriculum; duplicate name -> 409	**PASS**
Curricula	Create chapter; other instructor can list; no token -> 401	**PASS**
Workflow	New question is PENDING and PRIVATE by default; source MANUAL	**PASS**
Validation	Client cannot set status or owner on create	**PASS**
Validation	difficulty 7 -> 400; chapter from another curriculum -> 400; MCQ answer not in options -> 400	**PASS**
Privacy	Other instructor sees shared, not private	**PASS**
Privacy	?visibility=PRIVATE cannot reveal another instructor's questions	**PASS**
Privacy	Other instructor GET / approve / reject a private question -> 404	**PASS**
Workflow	Approve records approver + time; re-approve -> 409	**PASS**
Workflow	Owner rejects with reason; approving a rejected question -> 409	**PASS**
Bank	Bank contains only approved; ignores a status override in the query	**PASS**
Filtering	Filter by status + difficulty returns exactly the matching question	**PASS**
Input	Malformed JSON -> 400; empty body -> 400; malformed JSON on reject -> 400	**PASS**
Input	Password over 72 bytes -> 400	**PASS**
Database	Direct SQL: difficulty 9, estimatedTimeMinutes 0, chapter orderIndex 0 are rejected	**PASS**
Database	Direct SQL: APPROVED without approvedAt, and approvedAt on a non-approved question, are rejected	**PASS**
Database	Direct SQL: moving a question to a curriculum its chapter doesn't belong to is rejected	**PASS**
Rate limit	5 wrong passwords -> 401 each; 6th -> 429 with Retry-After; correct password also 429 while locked	**PASS**
Rate limit	Other accounts unaffected by one account's lock	**PASS**
Rate limit	Spoofed X-Forwarded-For (new fake IP per attempt) does not bypass the login limit	**PASS**
Rate limit	A successful login resets the failure counter	**PASS**
Rate limit	Wrong invite codes from spoofed IPs: 5 x 403, then 429; correct code also 429 while locked	**PASS**
Config	Server refuses to start with a JWT_SECRET under 32 characters (manual check)	**PASS**
Timing	Unknown email vs wrong password: 56 ms vs 55 ms average (manual check)	**PASS**

Rows group related assertions; the script reports 55 individual checks; the last two rows were verified manually. The final section intentionally trips the invite-code limit, so the dev server must be restarted before re-running within the window. The checks will be ported to a permanent automated suite (Vitest, isolated test database) in sub-step 5.

12. Engineering log, risks and open questions

12.1 Problems encountered and how they were resolved

Problem	Root cause	Resolution
npm ENOTEMPTY during install	Dependency install started while create-next-app's own install was still writing node_modules	Waited for all npm processes to exit, then reinstalled sequentially
Prisma P1010 'access denied'	A Homebrew PostgreSQL already listened on localhost:5432, so connections hit it instead of the container	Mapped the container to host port 5434 (5433 used by another project), configurable via DB_PORT
'docker compose' unknown	Machine has the standalone docker-compose binary (Colima), not the CLI plugin	npm scripts use docker-compose
.env.example ignored by git	Default Next.js .gitignore ignores .env*	Added !.env.example exception
prisma migrate dev refused the enum change	Removing an enum value that existing rows use needs interactive confirmation	Wrote the migration by hand (Section 4.6), applied with migrate deploy, verified with migrate diff
Rate limit bypassable by faking X-Forwarded-For (found by the new test)	The first design keyed login failures by IP + email, reading the IP from X-Forwarded-For. The Next.js server passes the client's header through unchanged, so a fresh fake IP per attempt reset the counter	Login limit keyed per account; client IPs trusted only with TRUST_PROXY=true behind a proxy that sets the header. The regression test now passes
Broken JSON caused HTTP 500	req.json() throws on malformed input and the error reached the generic handler	readJson() helper returns 400; used by every route
DB outage reported as 401	A try/catch around both token verification and the user lookup swallowed database errors	Narrowed the try/catch to token verification only; DB errors now surface as 500

12.2 Known limitations (to address in later sub-steps)

- **Rate-limit counters are in memory.** They reset when the server restarts and are not shared between instances; a multi-instance deployment needs a shared store such as Redis.
- **Per-account login lockout can be abused** to lock a known account for 15 minutes. Acceptable for a small invite-only user base; CAPTCHA or progressive delays are the usual next step.
- **Without TRUST_PROXY**, all invite-code attempts share one counter, so repeated wrong codes from anyone pause registration for everyone for 15 minutes.
- **Tokens cannot be revoked early.** Mitigated by the per-request user reload (deleted users apply immediately); a token denylist or short-lived access + refresh tokens can follow.
- **A single shared invite code** gates registration; per-person invites or an admin approval step would allow revoking one person's invite.
- **options JSON is validated by Zod, not by the database.** Acceptable for now; could move to a dedicated table if per-option analytics are needed.

12.3 Open design questions

- Should an instructor be allowed to approve their **own shared** question, or should shared questions require a second instructor (four-eyes principle)?
- Should a rejected question be **editable and resubmittable** (REJECTED -> PENDING), and should edits to an approved question send it back for review?
- Should instructors be able to change visibility after approval (e.g. share a previously private question)?
- Should rejections record **who rejected** the question and when? Today only approvals store the reviewer (approvedBy / approvedAt); a rejection stores only its reason. A generic reviewedBy / reviewedAt pair would close this gap.

13. Next steps

Sub-step	Deliverable
5 - Automated tests	Vitest suite on a separate test database covering the approval workflow and every privacy rule in Section 6
6 - Cloud readiness	Multi-stage Dockerfile (Next.js standalone output), docker-compose with app + db, README
Later steps	AI question generation (source = AI, still PENDING until approved); instructor-only AI exam generation by curriculum, chapters, difficulty and total duration